



CHEMISTRY (H2)
Paper 3 Free Response

9647/03
16th September 2015
2 hours

Candidates answer on separate paper.

Additional materials: Answer paper
Graph Paper
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, Civics Group, Centre number and Index number in the spaces provided on the cover page and on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a HB pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, glue or correction fluid.

Answer any **four** questions.

A Data Booklet is provided.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for good English and clear presentation in your answers.

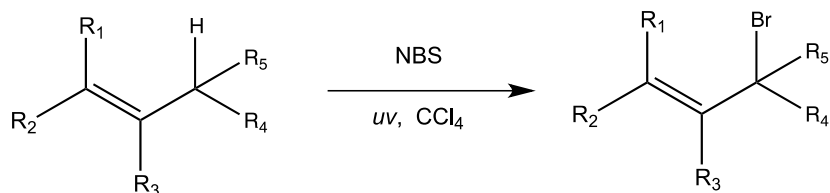
At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of 10 printed pages.

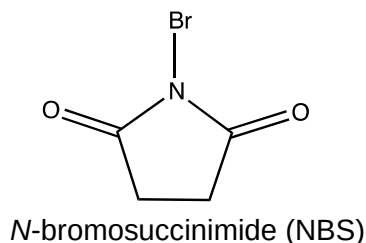
- 1 (a) The halogens and their compounds are useful laboratory reagents and have many applications.
- (i) Describe the trend in the colour of the halogens.
 - (ii) Explain, in terms of structure and bonding, the trend in the volatility of the halogens.
 - (iii) Halogens are oxidising agents.
By quoting appropriate values from the *Data Booklet*, explain the reactions of the halogens with sodium thiosulfate. Write balanced equations for the reactions.
- [6]
- (b) Hydrocarbons react with halogens under different conditions to give a wide range of organic compounds.

The Wohl-Ziegler bromination is shown below.



In this reaction, a bromine atom is incorporated into a molecule at the position next to a carbon-carbon double bond.

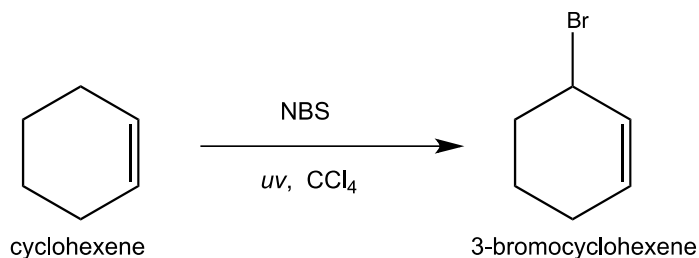
The reagent used for the reaction is *N*-bromosuccinimide (NBS).



The catalytically active species is bromine, which is present in trace amount in NBS samples. Subsequent amount of bromine is generated in situ when NBS reacts with hydrogen bromide.

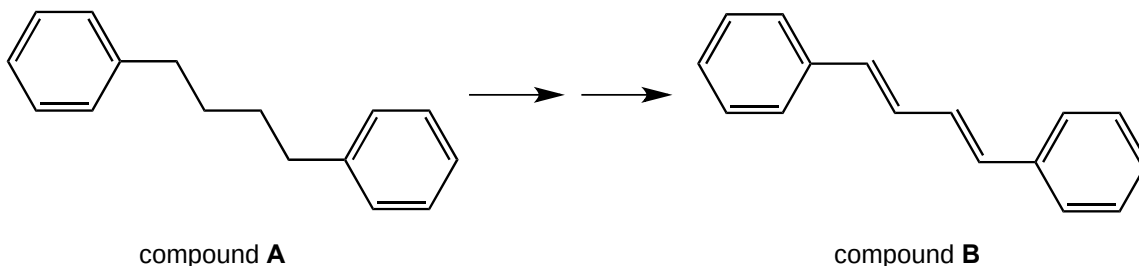
- (i) NBS reacts with hydrogen bromide in a 1:1 molar ratio to produce bromine and succinimide as the only products. Write a balanced equation for the reaction.

In one such reaction, cyclohexene is converted to 3-bromocyclohexene using NBS.



- (ii) Describe the mechanism for the formation of 3-bromocyclohexene from cyclohexene and bromine.
- (iii) State the type(s) of stereoisomerism that could be exhibited by 3-bromocyclohexene and draw the isomers.

In another reaction, compound **A** can be converted to compound **B** in two steps. One of the steps is the Wohl-Ziegler bromination.



(iv) Suggest suitable reagents and conditions and the structural formula for the intermediate formed.

(v) State the total number of stereoisomers of compound **B** and draw the isomers.

[10]

(c) The mineral fluorspar, which is mainly calcium fluoride, is a major source of fluorine.

The first stage in liberating the fluorine from calcium fluoride is to grind this compound up and react it with concentrated sulfuric acid. The products are hydrogen fluoride and calcium sulfate.

(i) Define the *standard enthalpy change of formation*, $\Delta H_f^\ominus$, of a substance.

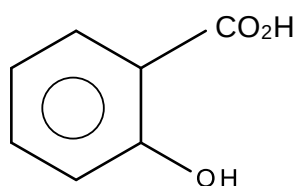
(ii) Using the following data of ΔH_f and lattice energy, construct an energy cycle to calculate the enthalpy change for the reaction between calcium fluoride and concentrated sulfuric acid.

$$\begin{aligned}\Delta H_f [\text{HF(g)}] &= -271 \text{ kJ mol}^{-1} \\ \Delta H_f [\text{CaSO}_4(\text{s})] &= -1434 \text{ kJ mol}^{-1} \\ \Delta H_f [\text{Ca}^{2+}(\text{g})] &= +1918 \text{ kJ mol}^{-1} \\ \Delta H_f [\text{F}^-(\text{g})] &= -249 \text{ kJ mol}^{-1} \\ \text{L.E.} [\text{CaF}_2(\text{s})] &= -2640 \text{ kJ mol}^{-1} \\ \Delta H_f [\text{H}_2\text{SO}_4(\text{l})] &= -814 \text{ kJ mol}^{-1}\end{aligned}$$

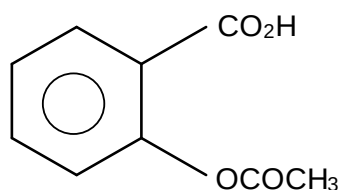
[4]

[Total: 20]

2 (a) In folk medicine, willow bark teas were used as headache remedies and other tonics. Its analgesic property is due to salicylic acid. Nowadays, salicylic acid is administered in the form of aspirin which is less irritating to the stomach.



Salicylic acid



Aspirin

(i) In the preparation of aspirin from salicylic acid, water is added to precipitate aspirin. Explain why aspirin is not soluble in water.

- (ii) Aspirin has a pK_a of 3.48 while benzoic acid has a pK_a of 4.2. Explain the difference in their pK_a values.
- (iii) Draw the structure of the 2 organic compounds formed when aspirin is heated with $NaOH(aq)$.

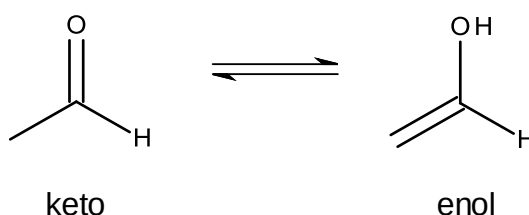
(iv)

Acid	$K_{a1} / \text{mol dm}^{-3}$	$K_{a2} / \text{mol dm}^{-3}$
Salicylic acid	1.07×10^{-3}	1.82×10^{-14}
Ethanoic acid	1.80×10^{-5}	-
Carbonic acid	4.50×10^{-7}	-

Using the information above, suggest the organic products formed if CO_2 is bubbled through the mixture formed in (a)(iii).

[8]

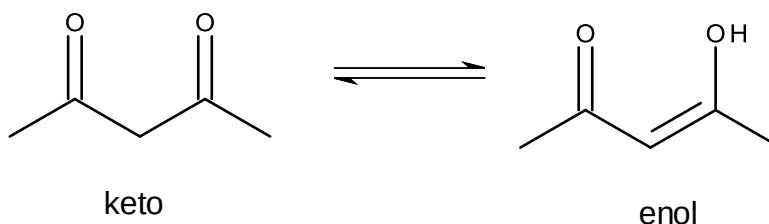
- (b) Tautomers are isomers with the same carbon skeleton and differ only in the positions of hydrogen atoms and electrons. Carbonyl compounds such as ethanal exhibits keto-enol tautomerisation as shown below.



Although these 2 isomers exist in equilibrium, the enol form is less stable than the keto form.

- (i) Using data from the Data Booklet, calculate the enthalpy change for the conversion of the keto to enol.

β -diketones such as pentan-2,4-dione also exhibits tautomerisation.

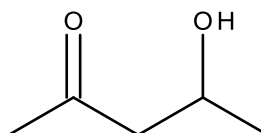


It is found that the enol form is more stable than the keto form.

- (ii) Suggest a reason for this.

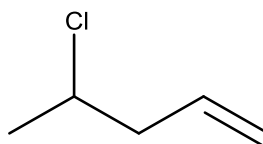
[3]

- (c) 4-hydroxypentan-2-one gives off a fresh herb smell and has the following structure.



4-hydroxypentan-2-one

- (i) 4-hydroxypentan-2-one can be synthesised from 4-chloropent-1-ene in 3 steps.



4-chloropent-1-ene

Suggest the reagents and conditions used in each step and the structural formula for the intermediate formed.

- (ii) Compound **R**, $C_6H_{12}O_2$, is neutral and does not decolourise $Br_2(aq)$. When compound **R** is heated with $NaOH(aq)$, one of the products obtained is HCO_2Na .

Upon refluxing compound **R** with acidified $KMnO_4$, compound **S** and carbon dioxide gas were formed. Compound **S** is an isomer of 4-hydroxypentan-2-one. It is chiral and gives effervescence with $Na_2CO_3(s)$.

Suggest the structures of compounds **R** and **S**, explaining your answer.

[9]

[Total: 20]

- 3 (a) Acid rain has harmful effects on aquatic animals, plants and infrastructure. It contains sulfuric acid formed from sulfur trioxide. Sulfur trioxide is produced upon atmospheric oxidation of sulfur dioxide from coal burning.

- (i) Write an equation for the formation of sulfuric acid from sulfur trioxide.

Even before the existence of acid rain, unpolluted rain water was slightly acidic due to dissolved CO_2 . The solubility of *pure* carbon dioxide gas in water is 88 cm^3 per 100 cm^3 of water under room conditions.

- (ii) Given that air contains 0.033% by volume of carbon dioxide, calculate the concentration in mol dm^{-3} of carbon dioxide dissolved in unpolluted rain water.

Dissolved carbon dioxide forms carbonic acid, H_2CO_3 , in water. The acid dissociation constant of carbonic acid is given below.



- (iii) Using your answers in (a)(ii), calculate the pH of unpolluted rain water (ignore the effect of K_{a2} on the pH).

Some fishes and shellfish die at pH values of 4.5 to 5.0. Lakes with limestone-rich soil can maintain a relatively stable pH even when acid rain falls due to the HCO_3^-/CO_3^{2-} buffer system.

- (iv) With aid of an equation, explain how lakes with limestone-rich soil maintain a relatively stable pH.

An environmental chemist needs a carbonate buffer of pH 10.0 to study the effects of acid rain on limestone-rich soils.

- (v) The acid dissociation constant, K_a , of HCO_3^- is $4.7 \times 10^{-11} \text{ mol dm}^{-3}$. Calculate the mass of solid Na_2CO_3 the chemist needs to add to 500 dm^3 of 0.2 mol dm^{-3} NaHCO_3 to form a buffer of pH 10.0.

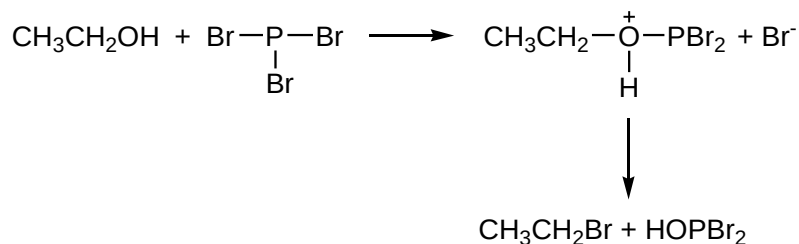
[9]

Oxides and halides of Period 3 elements have many applications.

- (b) A farmer intends to add a magnesium-containing compound to his farmland to correct magnesium deficiency and raise soil pH.
- (i) Suggest whether the farmer should add magnesium oxide or magnesium chloride to his farmland.
- (ii) Explain your answer in (b)(i) by describing the reactions of the compound suggested with water and acid. Write equations where appropriate and state the pH of the solution formed when the compound is dissolved in water.

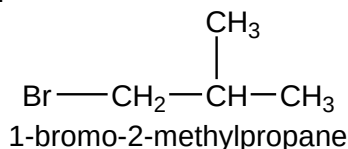
[4]

- (c) Phosphorus tribromide can be used to synthesise bromoethane from ethanol as shown below.



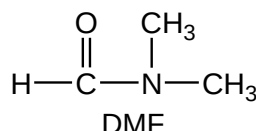
- (i) Name the reaction for the conversion of ethanol to bromoethane.
- (ii) Suggest how the reactivity of ethanol is increased in the presence of PBr_3 .
- (iii) Suggest why water cannot be used as a solvent for this reaction.
- (iv) Draw a structure to illustrate the shape about the P atom in a molecule of the product, HOPBr_2 . Suggest the shape about the P atom.

Both bromoethane and 1-bromo-2-methylpropane are primary alkyl halides that undergo S_N2 reaction with CN^- nucleophile.



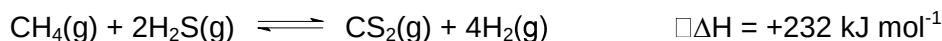
(v) Explain the following observations.

- Bromoethane undergoes S_N2 reaction with CN^- nucleophile at a faster rate than 1-bromo-2-methylpropane.
- Both compounds undergo S_N2 reactions with CN^- at a slower rate when the solvent is ethanol, CH_3CH_2OH , compared to using DMF, as solvent.



[7]
[Total: 20]

- 4 (a) Hydrogen sulfide is a colourless gas with a characteristic 'rotten egg' smell. It occurs naturally in crude petroleum and reacts with methane according to the following equilibrium.



- State and explain the effect on the number of moles of methane when the volume of the vessel is reduced at a constant temperature.
- Write an expression for the equilibrium constant, K_c , stating its units.
- In an experiment, 1.00 mol of CH_4 , 2.00 mol of H_2S , 1.00 mol of CS_2 and 2.00 mol of H_2 are mixed in a 250 cm^3 vessel at 960°C . The concentration of methane is found to be 5.56 mol dm^{-3} when the system reaches equilibrium. Calculate the value of K_c at 960°C .
- The equilibrium constant varies with temperature according to the van't Hoff equation:

$$\ln \frac{K_2}{K_1} = - \frac{\Delta H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

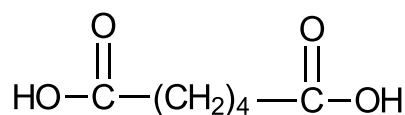
where K_1 is the equilibrium constant at temperature T_1
 K_2 is the equilibrium constant at temperature T_2
 $\Delta H^\ominus$ is the enthalpy change in J mol^{-1}
 R is the ideal gas constant and
 T is temperature in Kelvin.

Using your answer in (a)(iii), calculate the value of equilibrium constant at 480°C .

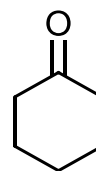
- Hydrogen sulfide can be removed by passing it through nitric acid. 0.15 mol of H_2S was found to react with 0.10 mol of HNO_3 to give a yellow solid and an oxide of nitrogen. Determine the oxidation state of nitrogen in the oxide. Hence, write a balanced equation for the reaction of H_2S and HNO_3 .

[10]

- (b) Adipic acid is one of the most important dicarboxylic acids and has a wide range of industrial applications. It is used as a monomer in the synthesis of nylon, and in the production of plasticizer and coatings.



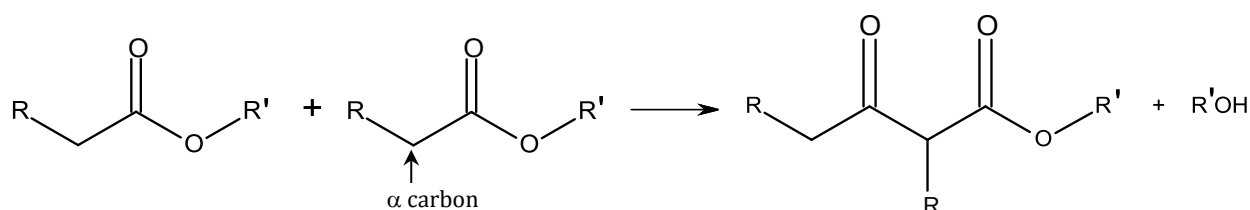
adipic acid



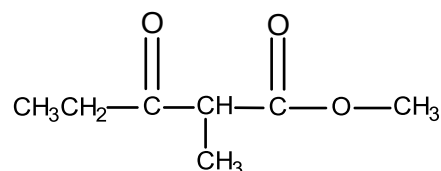
cyclohexanone

- (i) Show a three-step synthetic route to obtain adipic acid from cyclohexanone. Suggest reagents and conditions you would use and show the structures of any intermediates formed.
- (ii) Suggest a simple chemical test to distinguish between adipic acid and cyclohexanone. State the observations and write a balanced equation for any reaction that occurs.

In the 1880s, German chemist Rainer Ludwig Claisen discovered that esters were able to react in the presence of a strong base to give a keto ester. In the first step of the mechanism, a proton is removed from the α carbon of the ester molecule.

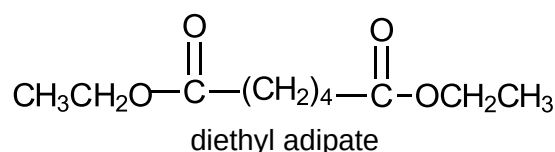


- (iii) Suggest a reason why the hydrogen atom on the α carbon is more acidic than the other hydrogen atoms in the same molecule.
- (iv) Name the type of reaction that has taken place between the ester molecules.
- (v) Two molecules of an ester undergoes the above reaction to give methanol and the following organic product.



Suggest the structure of the ester.

- (vi) Adipic acid undergoes esterification to form diethyl adipate.



The two ester functional groups of diethyl adipate can react intramolecularly to give a cyclic product and ethanol. Draw the structure of the cyclic product.

[10]
[Total: 20]

- 5 Aluminium chloride, AlCl_3 , exhibits properties which differ from chlorides of other Period 3 elements. It sublimes at a relatively low temperature of 180°C at atmospheric pressure due to the original lattice structure of AlCl_3 being converted into Al_2Cl_6 molecules. In the presence of excess water, aluminium chloride forms an acidic solution of pH 3 to 4.

(a) Describe and explain the reactions of aluminium chloride with excess water, writing equations where appropriate.

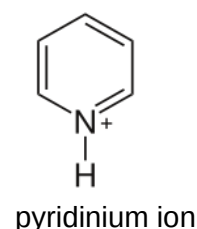
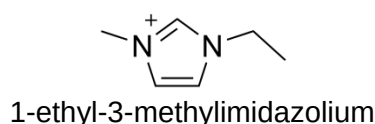
[2]

(b) Aluminium chloride reacts with sodium chloride to form sodium chloroaluminate, NaAlCl_4 . Sodium chloroaluminate is one of the simplest compounds containing chloroaluminate anion and has a melting point of 185°C .

(i) Explain why sodium chloroaluminate would be formed from the above reaction, stating the type of bond that is formed during this reaction.

(ii) Draw the dot-and-cross diagram of chloroaluminate anion. Using the valence shell electron pair repulsion theory, state its shape and bond angle.

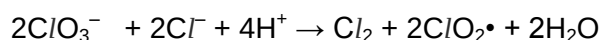
(iii) A class of compounds called room temperature ionic liquids (RTILs) can be formed from chloroaluminate anion and organic cations such as 1-ethyl-3-methylimidazolium and pyridinium ions.



Explain the differences in physical states between RTILs and sodium chloroaluminate in terms of bonding.

[6]

(c) To determine the rate equation of the following chlorate-chloride reaction, an experiment was conducted using $0.000480 \text{ mol dm}^{-3}$ of ClO_3^- , 0.1 mol dm^{-3} of Cl^- and 0.4 mol dm^{-3} of H^+ .



At regular 5-minutes intervals, small samples of the reaction mixture were withdrawn, quenched and placed into the UV-vis spectrometer to record its absorbance value. The absorbance value corresponds to the concentration of the product $\text{ClO}_2\cdot$.

(i) The results of the above experiment were shown below.

Time/min	0	5	10	15	20	25
Absorbance/A	0.000	0.211	0.348	0.436	0.494	0.531

Plot the graph of absorbance/A against time/min.

- (ii) Beer-Lambert's Law states that the absorbance values, A , is directly proportional to the concentration of absorbing species, c , as shown below.

$$A = \epsilon cl$$

where ϵ is the molar extinction coefficient and l is the path length, which is usually 1.0 cm.

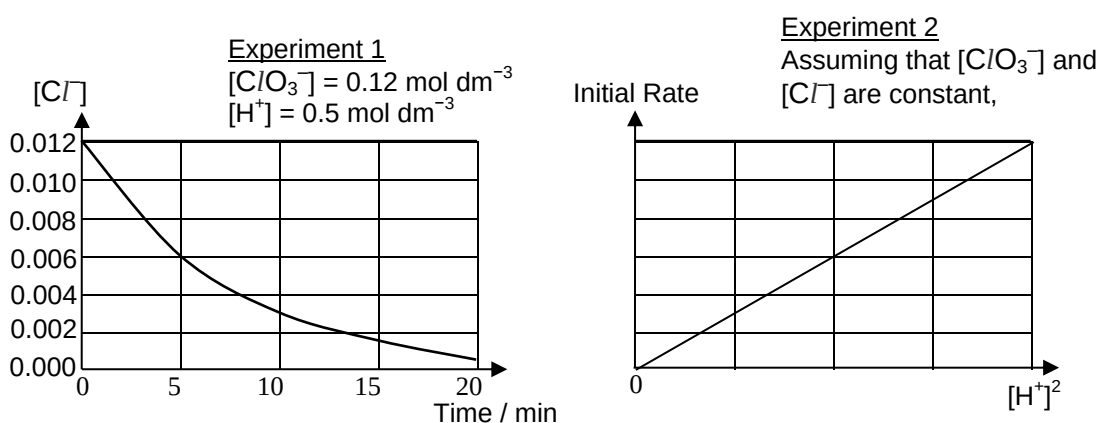
This equation can be used to calculate the absorbance value when maximum amount of $\text{ClO}_2^\bullet$ was formed from the above chlorate-chloride reaction.

Prove that the maximum absorbance value in the above experiment is 0.600, given that ϵ of $\text{ClO}_2^\bullet$ is $1250 \text{ mol}^{-1}\text{dm}^3\text{cm}^{-1}$. Show your working clearly.

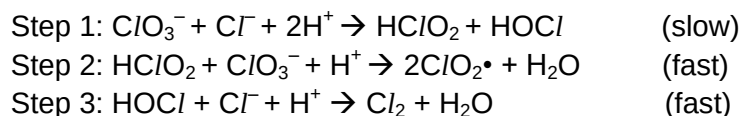
- (iii) Using your graph plotted in (c)(i) and the information given in (c)(ii), determine the half-life of the experiment and hence the order of reaction with respect to $[\text{ClO}_3^-]$.

A research student carried out a series of experiments to further investigate the order of reaction with respect to $[\text{Cl}^-]$ and $[\text{H}^+]$ in the chlorate-chloride reaction.

The following graphs were plotted.



- (iv) Using the above data, deduce the order of the reaction with respect to $[\text{Cl}^-]$ and $[\text{H}^+]$.
- (v) Hence, write a rate equation for the chlorate-chloride reaction.
- (vi) Using the half-life obtained from Experiment 1 in (c)(iv), calculate a value for the rate constant of the rate equation suggested in (c)(v).
- (vii) The following mechanism was proposed by a research student.



From the rate equation in (c)(v), deduce if the proposed mechanism is to be accepted or rejected.

[12]
 [Total: 20]